A

Major Project

On

DESIGN AND IMPLEMENTATION OF AN INTELLIGENT EMAIL AUTOMATION SYSTEM USING RPA AND GENERATIVE AI

(Submitted in partial fulfilment of the requirements for the award of the Degree)

BACHELOR OF TECHNOLOGY

In

INFORMATION TECHNOLOGY

By

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This is to certify that the project entitled "**DESIGN AND IMPLEMENTATION OF AN INTELLIGENT EMAIL AUTOMATION SYSTEM USING RPA AND GENERATIVE AI**" has being submitted by **K.Naina Harshitha (227R1A1230), S. Ganesh (227R1A1252), J. Adithya (227R1A1222)**, in partial fulfilment of the requirements for the award of the degree of B.Tech in Information Technology to the Jawaharlal Nehru Technological University Hyderabad, during the year 2025-26.

The results embodied in this project have not been submitted by any other university or institute for the award of degree or diploma.

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ABSTRACT

Email communication plays a vital role in both personal and organizational environments. With the increasing volume of emails received daily, manually reading, analyzing, and responding to emails has become time-consuming and inefficient. This project proposes an intelligent email automation system that integrates **Robotic Process Automation (RPA)** using **UiPath** with **Generative Artificial Intelligence (AI)** to automate email processing and response generation.

The system automatically retrieves incoming emails from the mail server using the **IMAP protocol** and extracts important information such as the email subject, body, and attachments. If attachments such as PDF, DOC, or Excel files are present, the system processes them and extracts relevant text. The extracted email content and attachment information are then summarized using **Generative AI**, which helps users quickly understand the key points of lengthy emails and documents.

To enhance usability and decision-making, the summarized content and reply suggestions are sent to the user through **WhatsApp Web automation** for review and approval. The system also performs **language detection and translation** to support multilingual email communication. After receiving the user's confirmation, the system automatically sends the final response email using the **SMTP protocol**.

The proposed system significantly reduces manual effort, improves response efficiency, and enhances email management by combining automation, artificial intelligence, and user-controlled decision-making. This solution provides a practical approach to intelligent email communication management in modern digital environments.

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1.INTRODUCTION

1.INTRODUCTION

Email communication plays an important role in both personal and organizational environments. Every day, individuals and organizations receive a large number of emails that require reading, understanding, and responding. Managing these emails manually can be time-consuming and may lead to delays, missed messages, or errors. Traditional email automation systems mainly rely on rule-based methods such as sorting emails, filtering spam, or sending predefined responses. However, these systems lack the ability to understand the actual meaning of email content and attachments. With the advancement of **Robotic Process Automation (RPA)** and **Generative Artificial Intelligence (AI)**, it is now possible to build intelligent systems that can process emails more effectively. RPA can automate repetitive tasks such as retrieving emails, extracting information, and sending responses, while Generative AI can analyze email content, summarize information, and generate meaningful replies.

This project proposes an **intelligent email automation system** that integrates UiPath RPA with Generative AI. The system automatically retrieves incoming emails, extracts the email body and attachments, and summarizes the content to reduce lengthy information. It also detects the language of the email and performs translation when necessary. The summarized information and reply suggestions are sent to the user through WhatsApp Web for review and approval before sending the final response.By combining automation, artificial intelligence, and user approval mechanisms, the proposed system reduces manual effort, improves response efficiency, and ensures reliable communication management.

1.1 PROJECT PURPOSE

The purpose of this project is to develop an intelligent email automation system that simplifies and improves the process of managing emails. In many organizations, handling a large number of emails manually can be time-consuming and inefficient. This project aims to reduce manual effort by using **Robotic Process Automation (RPA)** and **Generative Artificial Intelligence (AI)** to automatically read, analyze, and process incoming emails.

The system retrieves emails from the mail server, extracts the email content and attachments, and summarizes the information using Generative AI. It also detects the language of the email and performs translation when required. The summarized information and suggested replies are sent to the user through WhatsApp Web for review and approval. After the user confirms the response, the system automatically sends the reply through the email server.

1.2 PROJECT FEATURES

The proposed email automation system provides several features to improve the efficiency of email handling and communication management.

Automated Email Retrieval: The system automatically retrieves incoming emails from the mail server using the IMAP protocol.

Email Content Extraction: It extracts important information such as the email subject, body, and attachments for further processing.

Attachment Processing: The system reads text from attached documents such as PDF, Word, or Excel files.

AI-Based Summarization: Generative AI is used to summarize both the email content and the attachment text to reduce lengthy information.

Multilingual Support: The system detects the language of the email and translates it into the required language when needed.

Automated Reply Generation: Based on the summarized content, the system generates intelligent reply suggestions using Generative AI.

WhatsApp Notification and Approval: The summarized information and reply suggestions are sent to the user through WhatsApp Web for quick review and approval.

Automated Email Response: After user confirmation, the system automatically sends the reply using the SMTP protocol.

These features help reduce manual effort, improve response time, and ensure efficient email communication management.

2. LITERATURE SURVEY

2. LITERATURE SURVEY

Email automation has become an important area of research due to the increasing number of emails received daily in organizations and businesses. Several studies have focused on automating email processing using technologies such as Robotic Process Automation (RPA), Artificial Intelligence (AI), and Natural Language Processing (NLP). These systems aim to reduce manual effort and improve efficiency in handling emails. In many organizations, employees spend a significant amount of time reading emails, downloading attachments, categorizing messages, and responding to queries. Automating these repetitive tasks can improve productivity, reduce response time, and minimize human errors.

Robotic Process Automation (RPA) has been widely adopted to automate routine digital tasks such as email monitoring, data extraction, and automated responses. RPA tools like UiPath can interact with email servers, read incoming emails, extract information, and perform automated actions without requiring major changes to existing systems. Similarly, Artificial Intelligence techniques have been used to analyze email content, classify messages, and generate responses. Natural Language Processing (NLP) methods help computers understand human language, making it possible to analyze the meaning of emails and process text-based information effectively.

Khare et al. (2022) proposed an **E-Mail Assistant system using Robotic Process Automation**, which automatically reads incoming emails, classifies them, and organizes them into folders. The system also downloads email attachments and performs basic automation tasks using UiPath. However, the system mainly focuses on rule-based email management and does not provide intelligent content understanding or summarization.

Another study on **RPA-Driven Email Automation (2024)** presented a system that automatically detects spam and classifies emails based on predefined rules. The system filters unwanted emails and organizes them according to their categories. Although the approach reduces manual sorting of emails, it mainly focuses on filtering and classification rather than analyzing the actual meaning of email content.

An **AI-Powered Email Automation system (2025)** introduced the use of Artificial Intelligence with UiPath to process customer support emails. The system reads customer queries and generates automated responses using predefined workflows. While this approach improves response efficiency, it still relies on predefined reply patterns and lacks advanced features such as document summarization, multilingual processing, and user-controlled responses.

Recent advancements in Generative Artificial Intelligence have made it possible to go beyond traditional rule-based automation. Generative AI models can understand large amounts of text, summarize documents, detect languages, and generate meaningful responses based on context. These capabilities provide new opportunities for improving email automation systems by enabling intelligent content analysis and dynamic response generation.

From the literature review, it is observed that most existing systems focus on email classification, filtering, and automated replies using rule-based or workflow-based techniques. However, these systems do not fully utilize Generative AI for understanding email content and attachments, summarizing information, and generating intelligent responses with user approval. The proposed system addresses these limitations by integrating RPA with Generative AI to create an intelligent and user-controlled email automation solution.

2.1 REVIEW OF RELATED WORK

Several researchers have explored different approaches to automate email handling and management using technologies such as Robotic Process Automation (RPA), Artificial Intelligence (AI), and Natural Language Processing (NLP). These studies mainly focus on reducing manual effort, improving efficiency, and automating repetitive email-related tasks.

Khare et al. (2022) developed an **E-Mail Assistant system using Robotic Process Automation**, where UiPath bots automatically read incoming emails, classify them, and organize them into folders. The system also downloads attachments and performs basic email management tasks. Although the system helps in reducing manual work, it mainly relies on rule-based automation and does not include intelligent analysis or summarization of email content.

Another research work on **RPA-Driven Email Automation (2024)** proposed an automated system that detects spam and classifies emails using predefined rules. The system filters emails and organizes them into different categories based on specific conditions. While this method improves email sorting and filtering, it does not provide deeper understanding of the email content or generate intelligent responses.

An **AI-Powered Email Automation system (2025)** introduced the use of Artificial Intelligence with UiPath to handle customer support emails. The system reads customer queries and automatically generates replies using predefined workflows and templates. This approach improves response time for customer emails, but it still depends on predefined responses and lacks features such as document summarization, multilingual processing, and user approval mechanisms.

Recent developments in **Generative Artificial Intelligence** have opened new possibilities for intelligent email processing. Generative AI models can analyze large text data, summarize documents, detect languages, and generate meaningful responses. These capabilities allow automation systems to understand email content more effectively and provide context-aware responses.

From the review of related work, it can be observed that most existing systems focus mainly on rule-based email classification, spam filtering, and predefined automated replies. However, they lack advanced capabilities such as attachment summarization, multilingual support, and user-controlled response generation. The proposed system addresses these limitations by integrating UiPath RPA with Generative AI to create a more intelligent and interactive email automation system.

2.2 DEFINITION OF PROBLEM STATEMENT

In many organizations and businesses, a large number of emails are received every day. Managing these emails manually can be time-consuming and inefficient. Employees need to read each email, understand its content, check attachments, and prepare responses. This manual process often leads to delays in responding to important emails and increases the chances of human errors.

Existing email automation systems mainly rely on rule-based methods such as spam filtering, email classification, or predefined automated replies. These systems are limited because they do not fully understand the content of emails or the information present in attachments. In addition, many systems do not support multilingual communication or intelligent content summarization.

Therefore, there is a need for an intelligent email automation system that can automatically read emails, analyze their content and attachments, summarize the information, and generate meaningful responses. The system should also allow user approval before sending automated replies to ensure accuracy and reliability. The proposed project aims to address these limitations by integrating Robotic Process Automation (RPA) with Generative Artificial Intelligence to improve the efficiency of email handling and communication management.

2.3 EXISTING SYSTEM

Existing email automation systems mainly focus on automating basic email management tasks such as sorting emails, detecting spam, and sending predefined responses. These systems often use rule-based techniques or simple automation tools to reduce manual effort in handling emails. Robotic Process Automation (RPA) is commonly used to automate repetitive tasks like reading emails, organizing messages into folders, and extracting attachments.

In many existing systems, email classification and spam detection are performed using predefined rules or machine learning techniques. These systems help users filter unwanted emails and organize messages into different categories. Some systems also provide automated replies based on predefined templates or workflows to improve response efficiency.

Therefore, while existing systems help automate certain email management tasks, they do not provide a complete intelligent solution for understanding email content and generating context-aware responses. These limitations highlight the need for a more advanced system that integrates automation with Generative Artificial Intelligence to improve email processing and communication management.

Limitations of Existing System

Despite improvements in email automation technologies, the existing systems suffer from the following challenges:

•**Lack of intelligent content understanding**: Most systems rely on rule-based automation and cannot fully understand the contextual meaning of email content. They mainly perform basic tasks such as sorting emails or sending predefined replies.

•**Inability to process attachment content**: Many existing systems do not analyze the information present in email attachments such as PDF or document files, which often contain important details.

•**Limited response generation**: Automated replies are usually based on predefined templates or workflows, which may not generate accurate or context-aware responses.

•**Absence of user control and real-time interaction**: Existing automation systems often send replies automatically without user approval, which may lead to incorrect responses and reduced reliability.

2.4 PROPOSED SYSTEM

The proposed system introduces an intelligent email automation solution that integrates **Robotic Process Automation (RPA)** with **Generative Artificial Intelligence (AI)** to improve the efficiency of email handling and management. The system automatically retrieves incoming emails from the mail server and extracts important information such as the email subject, body, and attachments.

The extracted email content and attachment text are processed using Generative AI to generate concise summaries. This helps reduce the time required for users to read lengthy emails and documents. The system also detects the language of the received email and performs translation when necessary, enabling multilingual communication.

In addition, the system generates intelligent reply suggestions based on the summarized content of the email. These summaries and reply suggestions are sent to the user through **WhatsApp Web automation**, allowing the user to quickly review the information and provide approval.

After receiving user confirmation, the system automatically sends the final email response through the mail server. By combining automation, artificial intelligence, and user approval mechanisms, the proposed system reduces manual effort, improves response efficiency, and ensures accurate and reliable email communication management.

Advantages of the Proposed System:

The proposed system significantly improves upon existing email automation approaches by addressing several key limitations:

•**Intelligent Email Understanding**: The integration of Generative AI enables the system to understand the context of email content and attachments rather than relying only on predefined rules. This improves the accuracy of email processing.

• **Efficient Email Summarization**: The system summarizes lengthy email bodies and document attachments, allowing users to quickly understand important information without reading the entire content.

•**Multilingual Communication Support**: The system detects the language of incoming emails and performs translation when required, enabling communication across different languages.

•**Human-in-the-Loop Control**: The system sends summarized information and reply suggestions to the user through WhatsApp for approval before sending responses, ensuring accuracy and reliability.

•**Automated and Faster Response Generation**: Generative AI generates intelligent reply suggestions based on the summarized email content, helping users respond to emails more efficiently.

• **Improved Email Management Efficiency**: By combining RPA automation, AI-based analysis, and user approval mechanisms, the proposed system reduces manual workload and improves overall communication management.

2.5 OBJECTIVES

The main objectives of the proposed project are as follows:

• To **develop an intelligent email automation system** using Robotic Process Automation (RPA) and Generative Artificial Intelligence.

• To **automatically retrieve and process incoming emails** from the mail server.

• To **extract and analyze email content and attachments** for better understanding of the information.

• To **summarize email bodies and document attachments** using Generative AI to reduce lengthy content.

• To **detect and translate email languages** to support multilingual communication.

• To **generate intelligent reply suggestions** based on the summarized email content.

• To **implement a user approval mechanism** through WhatsApp before sending automated replies.

• To **improve efficiency and reduce manual effort** in email management and communication.

2.6 HARDWARE & SOFTWARE REQUIREMENTS

2.6.1 HARDWARE REQUIREMENTS:

Hardware interfaces specify the logical characteristics of each interface between the software product and the hardware components of the system. The following are the hardware requirements for implementing the proposed system:

•**Processor**	:	Intel Core i3 or higher
•**Hard Disk**	:	20 GB or more free space
•**RAM**	:	4 GB minimum

2.6.2 SOFTWARE REQUIREMENTS:

Software requirements specify the logical characteristics of each interface and the software components required to develop and run the proposed system. The following are the software requirements for the system:

• **Operating System**	:	Windows 10 / Windows 11
• **Development Tool**	:	UiPath Studio Community Edition
• **Framework**	:	UiPath Robotic Process Automation
• **Web Browser**	:	Google Chrome / Microsoft Edge

3. SYSTEM ARCHITECTURE & DESIGN

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Project architecture refers to the structural framework and design of the proposed email automation system, which includes its components, interactions, and overall workflow. It provides a clear structure for integrating UiPath Robotic Process Automation with Generative Artificial Intelligence to automate email processing and response generation. The architecture defines how emails are retrieved, processed, summarized, and how responses are generated with user approval.

3.1 PROJECT ARCHITECTURE

The project architecture defines the overall structure and workflow of the email automation system using UiPath RPA and Generative AI.

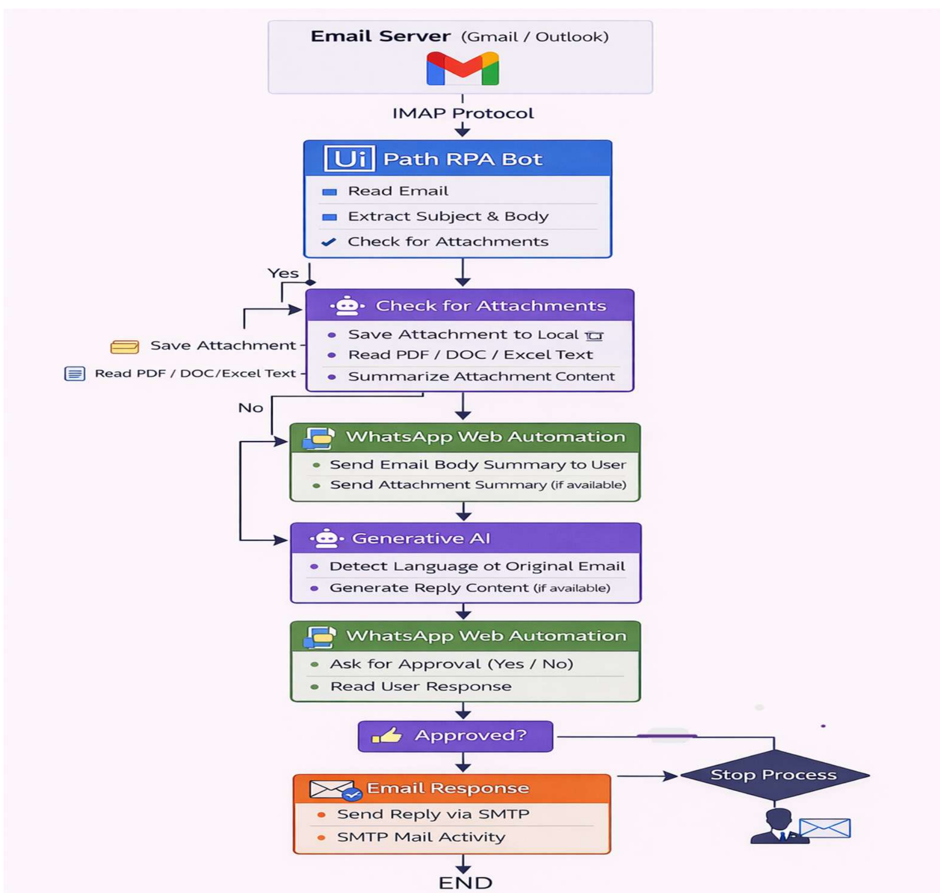

Figure 3.1: Project Architecture of Design and Implementation of an Intelligent Email Automation System Using RPA and Generative AI

3.2 DESCRIPTION

Input Data : The system processes incoming emails received from email servers such as Gmail or Outlook. The input data includes the email subject, body content, and attachments such as PDF, DOC, or Excel files.

Reading Data : The UiPath RPA bot retrieves emails using the IMAP protocol and extracts the email subject, body text, and attachment information for further processing.

Attachment Processing : If attachments are present, they are saved locally and the system reads the text from documents such as PDF, DOC, or Excel files to extract relevant information.

Content Summarization : The extracted email content and attachment text are processed using Generative Artificial Intelligence to generate concise summaries of the information.

Language Detection and Translation : The system detects the language of the incoming email and performs translation if the email is written in a different language.

Reply Generation : Based on the summarized content, Generative AI generates intelligent reply suggestions to assist the user in responding to emails efficiently.

User Interaction : The summarized information and reply suggestions are sent to the user through WhatsApp Web automation, where the user can review the content and provide approval.

Response Automation : After receiving user confirmation, the system automatically sends the final response through the email server using the SMTP protocol, completing the automated email management process.

3.3 UNIFIED MODELING LANGUAGE(UML)

Unified Modeling Language (UML) is not just a collection of diagrams but a powerful visual tool used to represent different aspects of a system. In this project, UML is used to design and understand the **Email Automation System using RPA and AI** from multiple perspectives.

UML helps in representing the workflow of the system, including how emails are received, processed, summarized, and responded to automatically. It provides a clear visualization of system components such as user interaction, email server communication, AI processing, and automated reply generation.

In this project, UML diagrams act as a **communication bridge** between developers and non-technical stakeholders. They simplify complex automation processes into easy-to-understand visuals, making it easier to explain how the system works.

3.3.1 USECASE DIAGRAM

A Use Case Diagram in Unified Modeling Language (UML) is a behavioral diagram that represents the functional requirements of a system by showing how different users, called actors, interact with the system to achieve specific goals, known as use cases. It provides a high-level visual overview of what the system does without explaining how it works internally. The diagram includes actors, use cases, and the relationships between them, helping to clearly illustrate system functionality and user interactions. It is mainly used during the requirement analysis phase to improve communication between developers, designers, and stakeholders by presenting system behavior in a simple and understandable way.

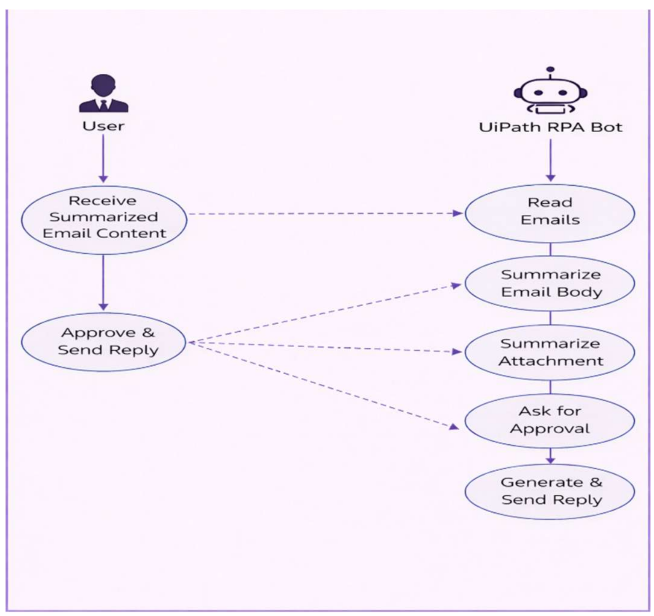

Figure 3.2 : Use Case Diagram

3.3.2 CLASS DIAGRAM

A Class Diagram in Unified Modeling Language (UML) is a type of static structure diagram that represents the structure of a system by showing its classes, along with their attributes, methods (operations), and the relationships between them. It provides a high-level view of the system's architecture by focusing on the main entities and how they interact with each other. Class diagrams help in identifying which class holds specific data and functionality, and they are mainly used during the design phase of software development to plan, visualize, and organize system components in a clear and structured way.

A **Class Diagram** in UML is a diagram that shows the structure of a system by representing classes, their properties, functions, and how they are connected to each other. It helps in understanding how different parts of a system are organized and interact, making it easier to design and develop software in a clear and systematic way.

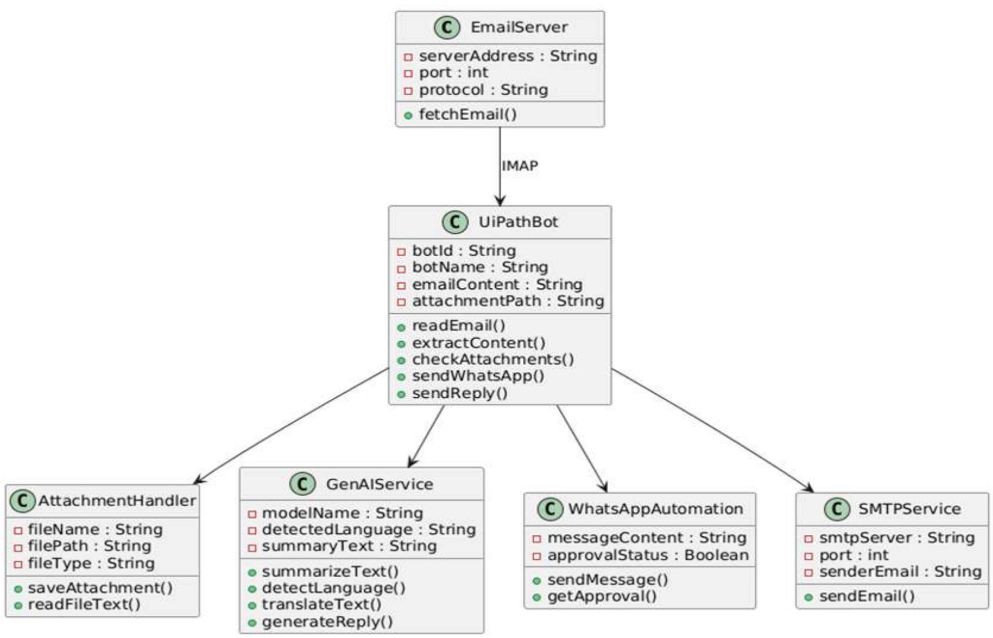

Figure 3.3: Class Diagram

3.3.3 SEQUENCE DIAGRAM

A **Sequence Diagram** in Unified Modeling Language (UML) is a type of behavioral diagram that shows how different objects or components of a system interact with each other in a specific sequence of time. It illustrates the flow of messages between objects, helping to understand how a particular process or functionality is carried out step by step. The diagram mainly focuses on the order of interactions, making it useful for analyzing system behavior during execution.

In a sequence diagram, objects are represented by vertical lines called lifelines, and interactions between them are shown using arrows that indicate messages passed from one object to another. It helps developers visualize the dynamic behavior of a system, identify the flow of control, and detect possible issues in communication between components during the design phase.

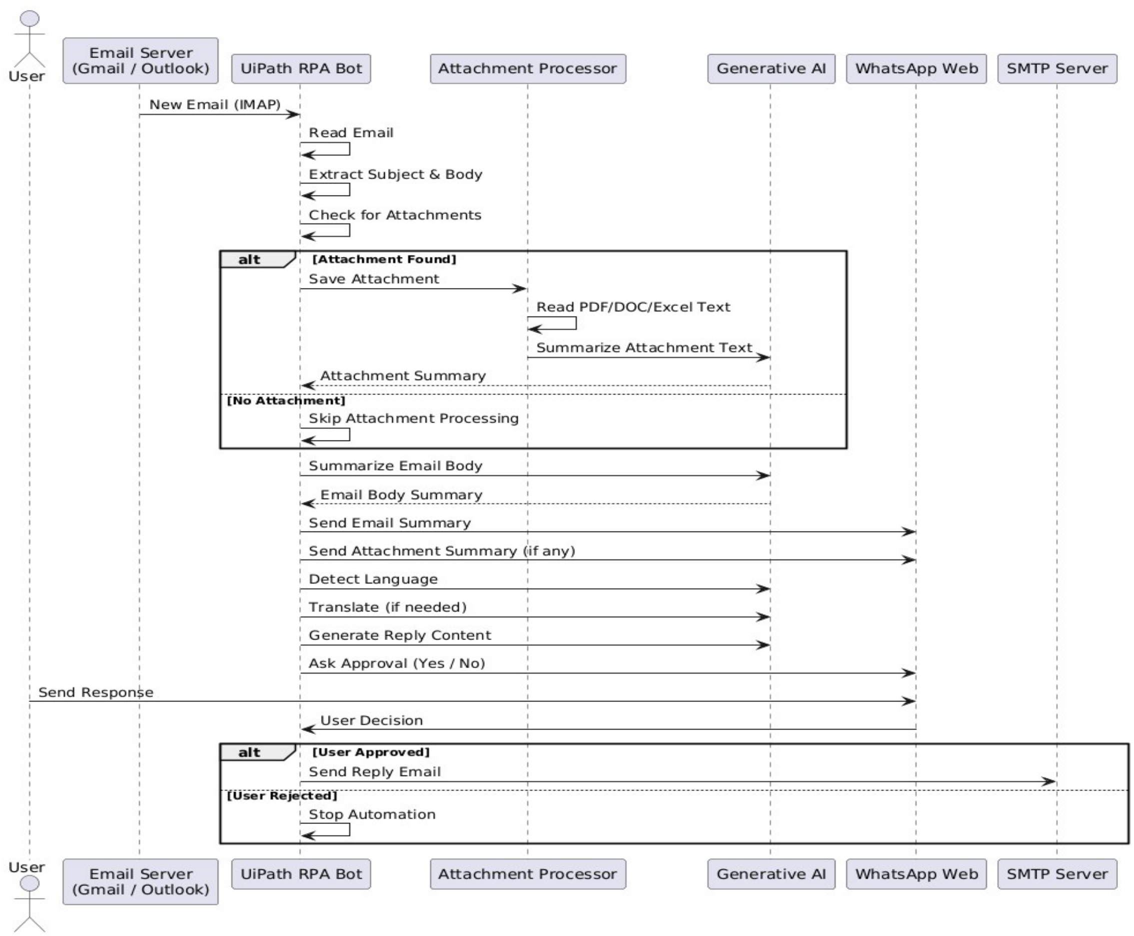

Figure 3.4: Sequence Diagram

3.3.4 ACTIVITY DIAGRAM

An activity diagram visually represents the workflow of a system, showing the sequence of actions and decision points. In our project, it illustrates how a user uploads a coin image, the system preprocesses the image and extracts features using a Convolutional Neural Network (CNN) model, and then predicts the denomination of the coin. The predicted result is finally displayed to the user through the application interface.

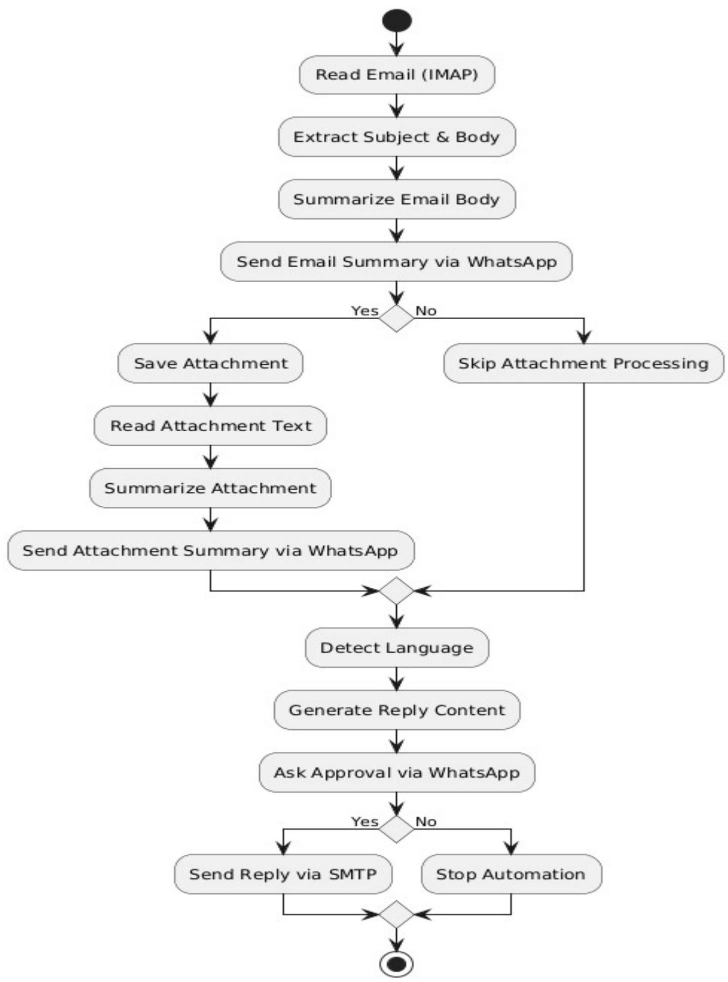

Figure 3.5 : Activity Diagram

3.3.5 DATA FLOW DIAGRAM

A Data Flow Diagram (DFD) is a graphical representation that illustrates how data moves within the proposed email automation system. It shows how data is received, processed, and transferred between different components of the system. The DFD helps in understanding the flow of information and the interaction between system components.

A Data Flow Diagram consists of four main elements:

•**External Entities:** Represent sources or destinations of data outside the system, such as the email server or the user.

•**Processes:** Indicate operations performed on the data, such as reading emails, summarizing content, and generating replies.

•**Data Flows:** Show the movement of data between different components of the system.

•**Data Stores:** Represent locations where data such as email content or attachments may be temporarily stored.

These components are represented using standard symbols such as circles for processes, arrows for data flows, rectangles for external entities, and open rectangles for data stores.

Benefits:

Data Flow Diagrams provide a clear visual representation of how information moves within the email automation system. They make it easier for both developers and users to understand how emails are received, processed, and responded to by the system. DFDs help in simplifying complex workflows, identifying possible issues in data movement, and improving the overall system design. They also support better communication among team members during the development process.

Applications:

DFDs are commonly used in system planning, software design, and process analysis. In this project, the diagram helps illustrate how emails are collected from the mail server, processed through the UiPath automation workflow, summarized using

Generative AI, and finally approved by the user before sending a response. This visualization helps in understanding the interaction between different system components.

Overall, a Data Flow Diagram plays an important role in designing and analyzing the email automation system

Levels:
DFDs are organized in different levels to represent the system in a structured manner.

• **Level 0 (Context Level):** Shows a general overview of the system and how it interacts with external entities such as the user and the email server.

• **Level 1:** Divides the main system into smaller processes like email retrieval, attachment processing, AI-based summarization, and approval handling.

• **Level 2:** Provides detailed steps within each process, explaining how individual operations such as language detection, reply generation, and email sending are performed.

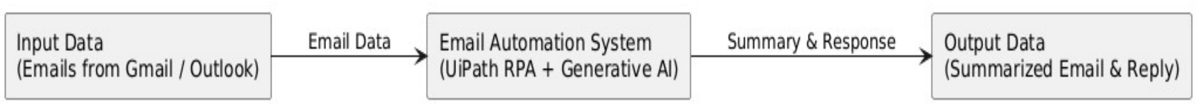

Figure 3.6: Dataflow Diagram of Design and Implementation of an Intelligent Email Automation System Using RPA and Generative AI

4. IMPLEMENTATION

4. IMPLEMENTATION

The implementation phase of the project focuses on developing and integrating the automation workflow to process emails efficiently. The system is implemented using **UiPath Robotic Process Automation (RPA)** combined with **Generative Artificial Intelligence (AI)** for intelligent email processing. During this phase, the designed architecture and workflows are executed to automate email retrieval, content extraction, summarization, and automated response generation. Proper monitoring and coordination of system components ensure that the system performs efficiently and meets the expected project objectives.

4.1 ALGORITHMS / TECHNIQUES USED

RPA-Based Automation for Email Processing

Robotic Process Automation (RPA) is used to automate repetitive tasks involved in email management. UiPath bots are designed to monitor incoming emails, retrieve messages from the email server, and process them automatically. The automation workflow interacts with email services through **IMAP and SMTP protocols** to read and send emails.

Advantages of RPA-Based Automation:

•Reduces manual effort in reading and responding to emails.
•Automates repetitive tasks such as email retrieval and response generation.
•Improves processing speed and operational efficiency.

Limitations of RPA-Based Automation:

•RPA alone cannot understand the contextual meaning of email content.
•Requires integration with AI technologies for intelligent decision-making.

Generative AI for Email Content Summarization

Generative Artificial Intelligence is used to analyze and summarize email content and attachments. The AI model processes the extracted email text and generates a concise summary that highlights the most important information.

Advantages of Generative AI:

•Automatically summarizes lengthy email content.

•Generates intelligent responses based on context.

•Improves productivity by reducing the time required to read emails.

Limitations of Generative AI:

•Requires proper input formatting for accurate results.

•Performance depends on the quality of the input data.

Language Detection and Translation

The system includes a language detection module that identifies the language of incoming emails. If the email is written in a different language, the system performs translation to ensure that the user can understand the content.

Advantages:

•Enables multilingual communication.

•Improves accessibility for users receiving emails in different languages.

User Approval Mechanism

To ensure reliability and prevent incorrect automated responses, the system includes a **human-in-the-loop mechanism**. The summarized email content and reply suggestions are sent to the user through **WhatsApp Web automation**. The user reviews the summary and decides whether to approve or reject the response.

Advantages:

•Improves reliability of automated responses.

•Prevents incorrect or inappropriate replies.

SMTP-Based Email Response Automation

Once the user approves the generated response, the system automatically sends the email reply using the **SMTP protocol**. This ensures that responses are delivered quickly and efficiently without manual intervention.

Modules of the Proposed System

To implement the email automation system, the following modules are designed:

1. **Email Retrieval Module**
   This module retrieves incoming emails from the mail server using the IMAP protocol.
2. **Email Content Extraction Module**
   Extracts the subject, body, and attachments from received emails.
3. **Attachment Processing Module**
   Reads text from attachments such as PDF, DOC, or Excel files.
4. **AI Summarization Module**
   Processes email content and attachments using Generative AI to generate summaries.
5. **User Interaction Module**
   Sends summarized content and reply suggestions to the user through WhatsApp for approval.
6. **Email Response Module**
   After user approval, the system sends the final email reply automatically using the SMTP protocol.

4.2 SAMPLE CODE

```
<Activity mc:Ignorable="sap sap2010"
 xmlns="http://schemas.microsoft.com/netfx/2009/xaml/activities"
 xmlns:ui="http://schemas.uipath.com/workflow/activities"
 xmlns:sap="http://schemas.microsoft.com/netfx/2009/xaml/activities/presentation"
 xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml">

  <Sequence DisplayName="Email Automation Main Workflow">

    <!-- Variables -->
    <Variable Name="getmail" Type="System.Collections.Generic.List(MailMessage)" />
    <Variable Name="mailBody" Type="String" />
    <Variable Name="mailSubject" Type="String" />
    <Variable Name="mailFrom" Type="String" />
    <Variable Name="pdfText" Type="String" />
    <Variable Name="attachmentContent" Type="String" />
    <Variable Name="emailSummary" Type="String" />
    <Variable Name="attachmentSummary" Type="String" />
    <Variable Name="finalSummary" Type="String" />
    <Variable Name="replyContent" Type="String" />
    <Variable Name="userChoice" Type="String" />

    <!-- Step 1: Get Email -->
    <ui:GetIMAPMailMessages MailFolder="Inbox"
```

```
                    OnlyUnreadMessages="True"
                    Top="1"
                    Result="[getmail]" />

  <!-- Step 2: Extract Data -->
  <Assign>
    <To>mailBody</To>
    <Value>[getmail(0).Body]</Value>
  </Assign>
  <Assign>
    <To>mailSubject</To>
    <Value>[getmail(0).Subject]</Value>
  </Assign>
  <Assign>
    <To>mailFrom</To>
    <Value>[getmail(0).From.ToString]</Value>
  </Assign>

  <!-- Step 3: Check Attachments -->
  <If Condition="[getmail(0).Attachments.Count > 0]">
    <Then>
      <ui:SaveAttachments Attachments="[getmail(0).Attachments]"
                    FolderPath="Attachments" />
      <ui:ReadPDFText FilePath="Attachments\file.pdf"
```

```
            Text="[pdfText]" />
    <Assign>
      <To>attachmentContent</To>
      <Value>[pdfText]</Value>
    </Assign>
  </Then>
  <Else>
    <Assign>
      <To>attachmentContent</To>
      <Value>""</Value>
    </Assign>
  </Else>
</If>

<!-- Step 4: Email Summarization -->
<ui:ConnectorActivity>
  <ui:ConnectorActivity.Parameters>
    <InArgument x:TypeArguments="x:String"
x:Key="prompt">[mailBody]</InArgument>
  </ui:ConnectorActivity.Parameters>
</ui:ConnectorActivity>
<Assign>
  <To>emailSummary</To>
  <Value>[mailBody]</Value>
</Assign>
```

```
<!-- Step 5: Attachment Summarization -->
<If Condition="[attachmentContent <> ""]">
  <Then>
    <ui:ConnectorActivity>
      <ui:ConnectorActivity.Parameters>
        <InArgument x:TypeArguments="x:String" x:Key="prompt">[attachmentContent]</InArgument>
      </ui:ConnectorActivity.Parameters>
    </ui:ConnectorActivity>
    <Assign>
      <To>attachmentSummary</To>
      <Value>[attachmentContent]</Value>
    </Assign>
  </Then>
</If>

<!-- Step 6: Combine Summary -->
<Assign>
  <To>finalSummary</To>
  <Value>"Email Summary: " + emailSummary +
         " Attachment Summary: " + attachmentSummary</Value>
</Assign>

<!-- Step 7: WhatsApp Message -->
<ui:OpenBrowser Url="https://web.whatsapp.com">
```

```
  <ui:TypeInto Text="[finalSummary]" />
</ui:OpenBrowser>

<!-- Step 8: User Input -->
<ui:InputDialog Label="Choose timing"
                Result="[userChoice]" />

<!-- Step 9: Generate Reply -->
<ui:ConnectorActivity>
  <ui:ConnectorActivity.Parameters>
    <InArgument x:TypeArguments="x:String"
x:Key="prompt">[finalSummary]</InArgument>
  </ui:ConnectorActivity.Parameters>
</ui:ConnectorActivity>
<Assign>
  <To>replyContent</To>
  <Value>[finalSummary]</Value>
</Assign>

<!-- Step 10: Switch -->
<Switch x:TypeArguments="x:String" Expression="[userChoice]">
  <Switch.Cases>

    <x:String x:Key="0 sec">
      <ui:SendSMTPMailMessage
```

```
      To="[mailFrom]"
      Subject="[mailSubject]"
      Body="[replyContent]"
      Server="smtp.gmail.com"
      Port="587"
      EnableSSL="True" />
  </x:String>
  <x:String x:Key="Few Hours">
    <Delay Duration="02:00:00" />
    <ui:SendSMTPMailMessage
      To="[mailFrom]"
      Subject="[mailSubject]"
      Body="[replyContent]"
      Server="smtp.gmail.com"
      Port="587"
      EnableSSL="True" />
  </x:String>
  <x:String x:Key="Tomorrow">
    <Delay Duration="24:00:00" />
    <ui:SendSMTPMailMessage
      To="[mailFrom]"
      Subject="[mailSubject]"
      Body="[replyContent]"
      Server="smtp.gmail.com"
```

```
       Port="587"
       EnableSSL="True" />
   </x:String>
  </Switch.Cases>
  <Switch.Default>
   <ui:LogMessage Level="Error"
          Text="Invalid Option Selected" />
  </Switch.Default>
 </Switch>
 <!-- Step 11: Loop Example -->
 <ForEach x:TypeArguments="MailMessage" Values="[getmail]">
  <ActivityAction x:TypeArguments="MailMessage">
   <Sequence>
    <Assign>
     <To>mailBody</To>
     <Value>[item.Body]</Value>
    </Assign>
    <Assign>
     <To>mailSubject</To>
     <Value>[item.Subject]</Value>
    </Assign>

   </Sequence>
  </ActivityAction>
```

```
    </ForEach>


    <!-- Step 12: Logging -->
    <ui:LogMessage Text="Workflow Started" Level="Info" />
    <ui:LogMessage Text="Email Retrieved" Level="Info" />
    <ui:LogMessage Text="Summarization Completed" Level="Info" />
    <ui:LogMessage Text="Reply Generated" Level="Info" />
    <ui:LogMessage Text="Email Sent Successfully" Level="Info" />


    <!-- Step 13: End -->
    <ui:LogMessage Text="Workflow Completed Successfully"
                   Level="Info" />
  </Sequence>
</Activity>
```

5. RESULTS & DISCUSSION

5. RESULTS & DISCUSSION

The following screenshots demonstrate the results of the proposed Email Automation System using UiPath Robotic Process Automation and Generative Artificial Intelligence. These screenshots illustrate the functioning of different modules in the system, including email retrieval, content summarization, user approval, and automated email response generation. The visual outputs provide a clear understanding of how the system processes incoming emails and generates automated replies efficiently.

5.1 UiPath Workflow Interface:

In the following screen, the main UiPath workflow is displayed. The workflow contains different automation activities such as email retrieval, content extraction, AI-based summarization, user approval processing, and SMTP email sending.

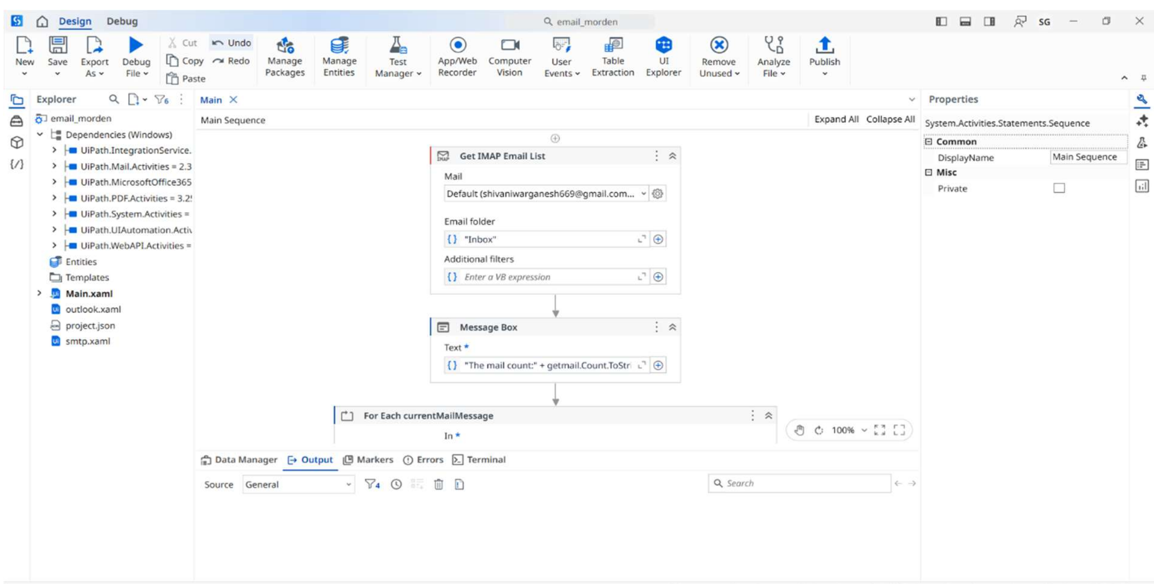

Figure 5.1: UiPath workflow interface of the Email Automation System.

5.2 Incoming Email Received

In this stage, the system receives an incoming email from the Gmail or Outlook server. The UiPath bot monitors the inbox and automatically retrieves new emails for processing.

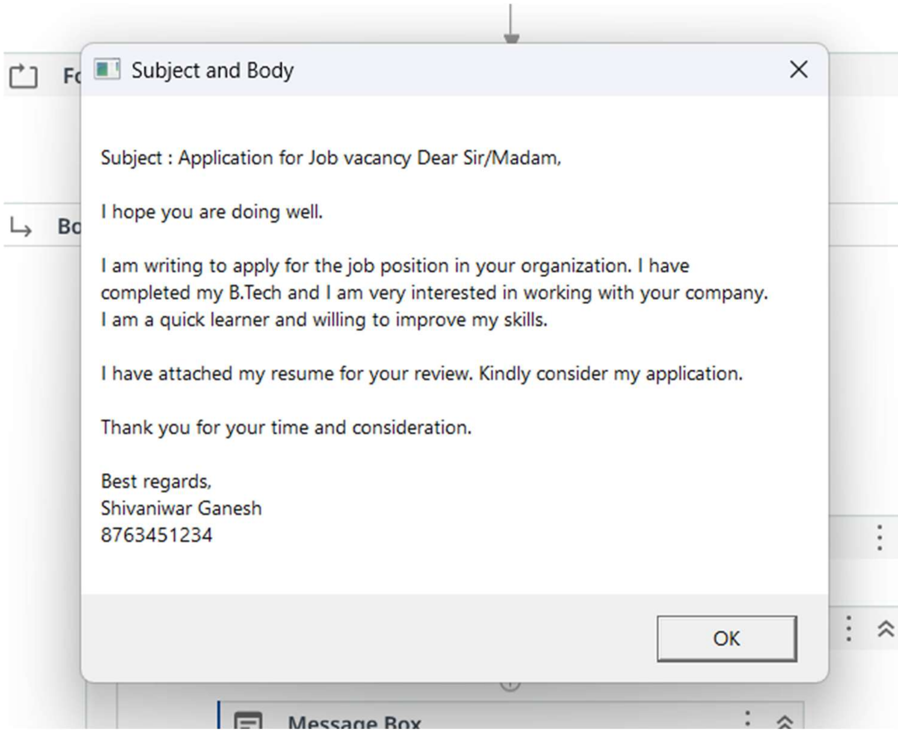

Figure 5.2: Extracted email content from the received message.

5.3 Attachment Text Extraction Result

If the email contains attachments such as PDF or document files, the system processes these files and extracts the textual information contained within them. This step allows the system to analyze the information present in attachments along with the email content. Extracting attachment text is important because many important details are often included in documents attached to emails.

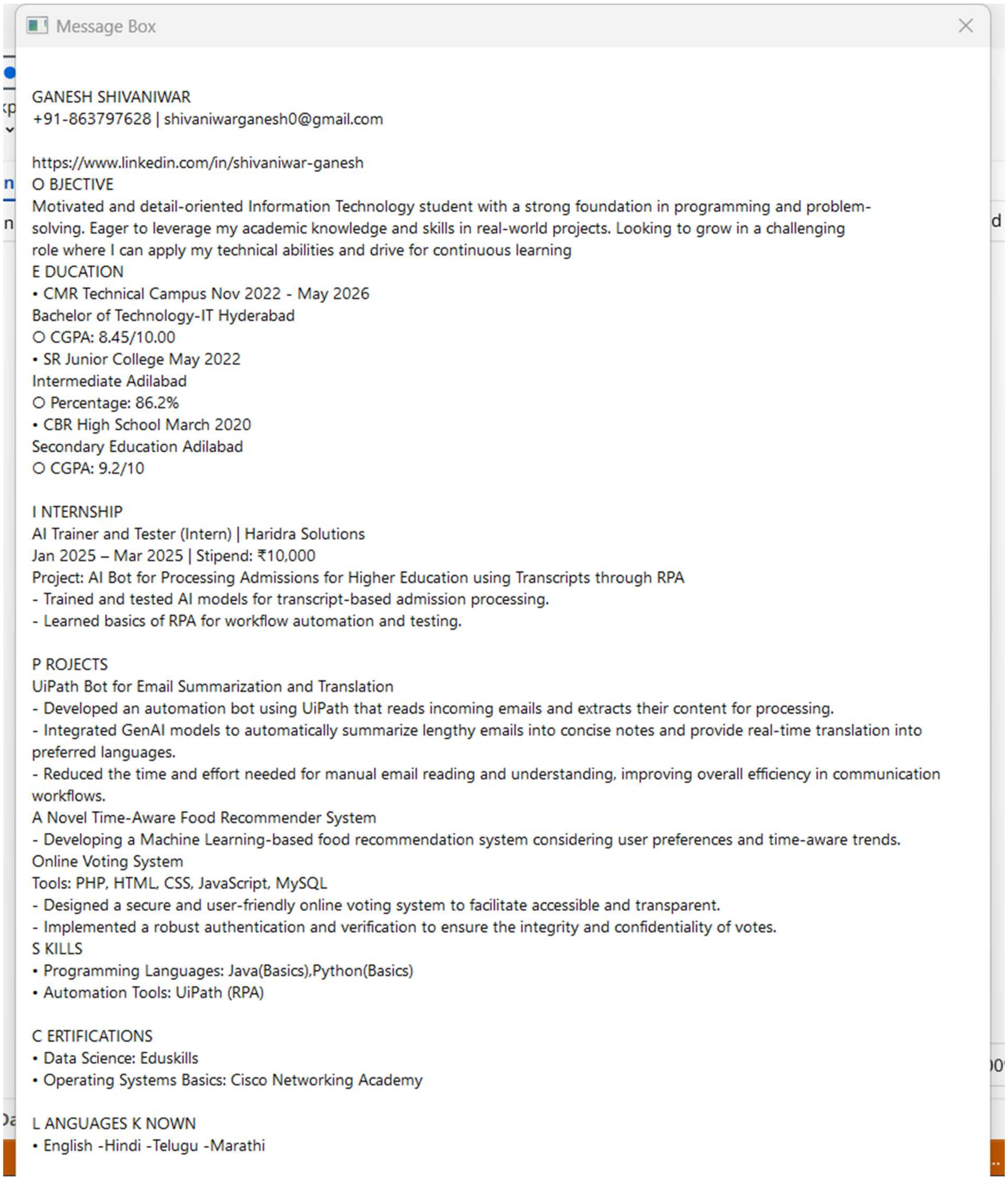

Figure 5.3: Extracted text from the email attachment.

5.4 AI Generated Email Summary

The extracted email content and attachment text are processed using Generative AI to generate a short summary. This helps the user quickly understand the important information in the email.

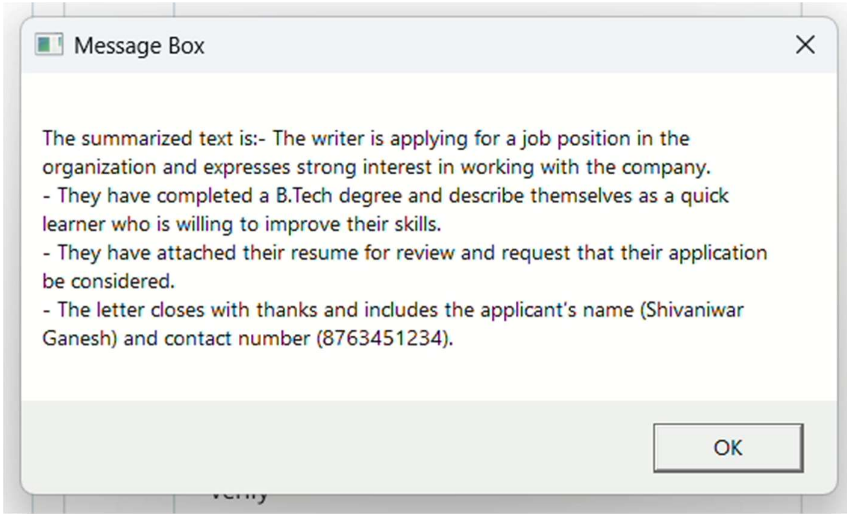

Figure 5.4: AI-generated summary of the email content.

5.5 WhatsApp Notification Sent to User

The summarized email content is sent to the user through WhatsApp Web automation. The user can review the summary and decide whether to approve the reply.

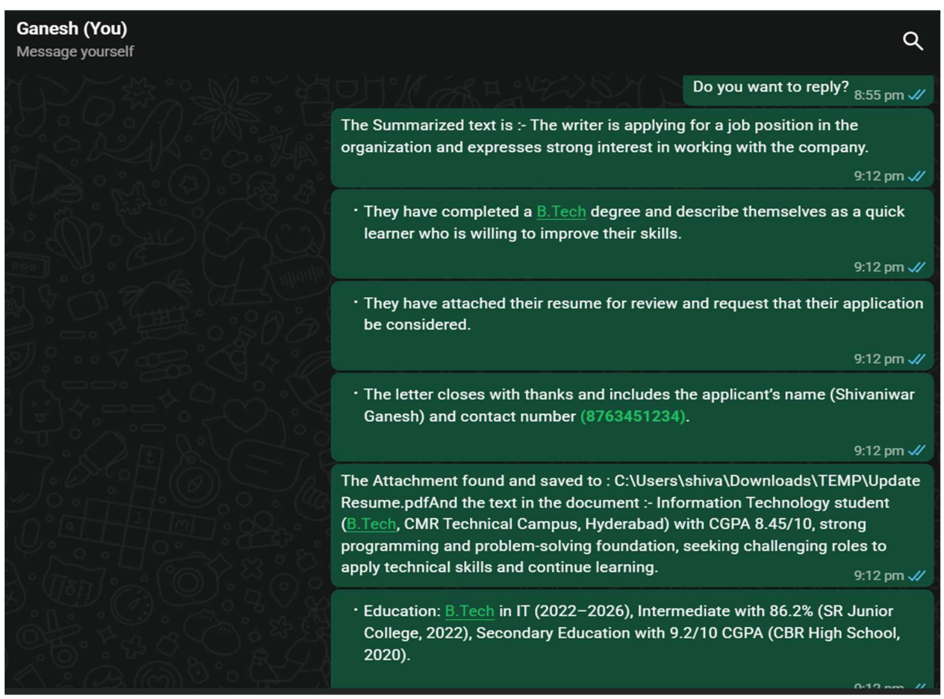

Figure 5.5: WhatsApp notification sent to the user.

5.6 Language Detection

In this stage, the system automatically detects the language of the received email using Generative AI. This helps the system understand emails written in different languages before processing them.

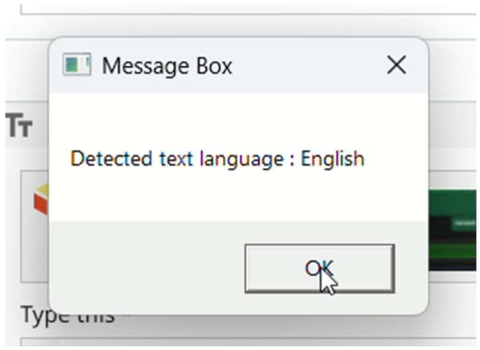

Figure 5.6: Detected language of the received email.

5.7 Email Translation

If the email is written in a language other than English, the system translates the content into English for better understanding and processing. This ensures that the system can handle multilingual email communication.

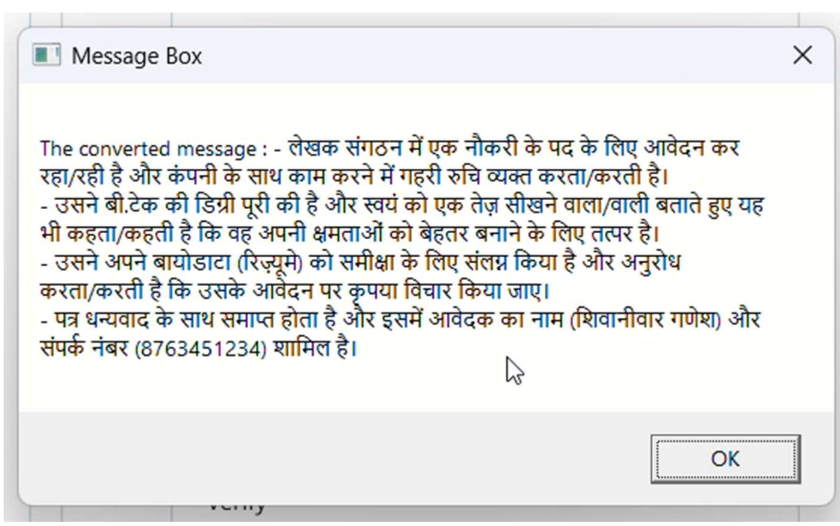

Figure 5.7: Translated email content generated by the system.

5.8 User Approval Response

In this step, the user provides a response to the approval request sent by the system. The user can either approve the reply, reject it, or choose a specific time for sending the reply. This step introduces a human-in-the-loop mechanism to ensure that automated responses are accurate and appropriate.

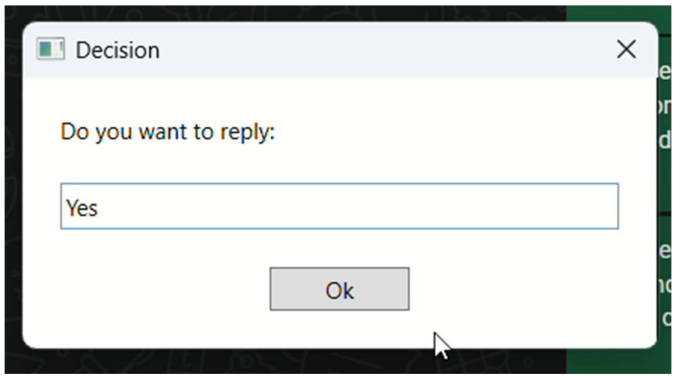

Figure 5.8: User approval received for sending the reply.

5.9 Generated Email Reply

Based on the summarized email content, the system generates a reply message using Generative AI. The generated response is designed to be professional and relevant to the context of the email. This automated reply generation helps users save time when responding to emails. Additionally, the system ensures that the reply is clear and appropriate for the email request. The generated response can also be reviewed and approved by the user before sending, which improves accuracy and reliability in automated communication.

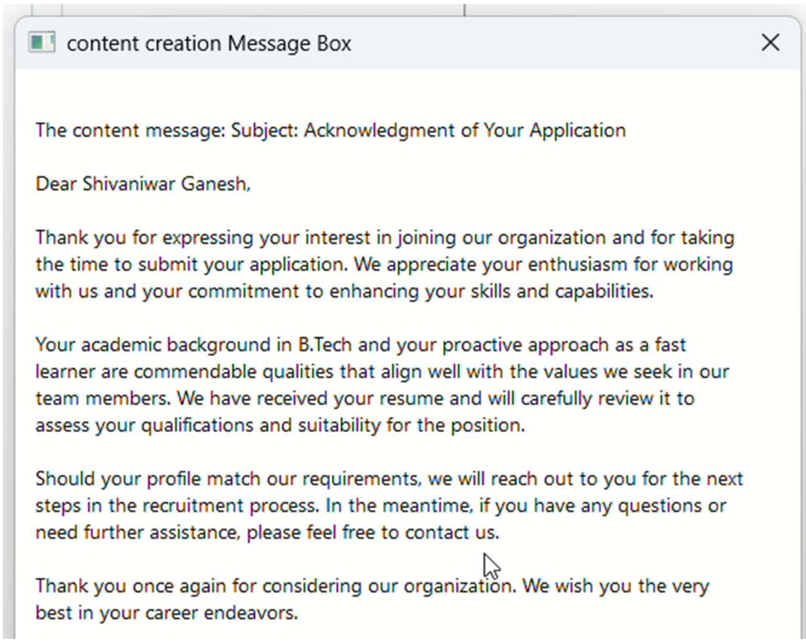

Figure 5.9: AI-generated reply based on email content.

5.10 Final Email Sent Successfully

After receiving user approval, the system automatically sends the reply email using the SMTP protocol. This completes the email automation process.

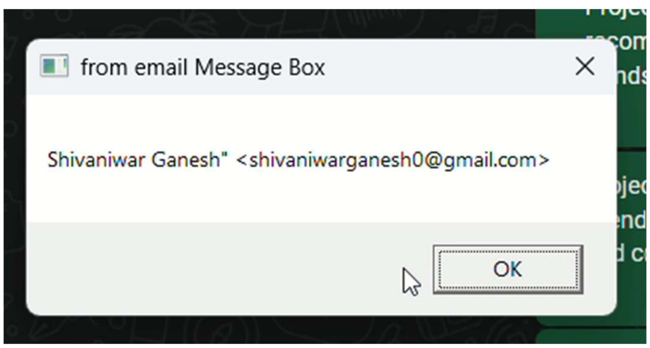

Figure 5.10: Automated reply email sent successful.

6. VALIDATION

6. VALIDATION

The validation of this project is carried out through systematic testing to ensure that the proposed Email Automation System using UiPath RPA and Generative Artificial Intelligence functions correctly and efficiently. The testing process evaluates different components of the system such as email retrieval, content extraction, AI-based summarization, user approval handling, and automated email response generation. Through well-defined test cases, the system is tested under various scenarios to verify its reliability and accuracy in processing incoming emails.

6.1 INTRODUCTION

The testing process begins by verifying whether the system can successfully connect to the email server and retrieve incoming emails using the **IMAP protocol**. Once the email is retrieved, the system extracts the subject, body, and attachments for further processing. This step ensures that the automation workflow correctly captures all necessary email information.

Next, the **Generative AI summarization module** is tested to confirm that it generates concise summaries from lengthy email content and attachments. The summarized information is then sent to the user through **WhatsApp Web automation**, allowing the user to review the message and provide approval before sending the response.

The system also verifies the **reply scheduling functionality**, where the user can choose to send the response immediately, after a delay, or on the next day using the switch case logic. This ensures that the automation behaves correctly under different timing conditions.

Finally, the **SMTP email sending module** is tested to confirm that the automated reply is successfully delivered to the recipient after receiving user approval. The results of these tests demonstrate that the system can reliably automate email processing, improve response efficiency, and reduce manual effort in managing email communications.

6.2 TEST CASES

TABLE 6.2.1 EMAIL RETRIEVAL TEST

Test case ID	Test case name	Purpose	Test Case	Output
1	Email Retrieval	To check whether the system retrieves incoming emails from the email server.	The system retrieves new emails from Gmail/Outlook using IMAP protocol.	Email successfully retrieved by UiPath bot.

TABLE 6.2.2 EMAIL SUMMARIZATION TEST

Test Case ID	Test Case Name	Purpose	Input	Output
1	Body Summarization	To check whether the system can summarize the email content	Email body text	Generated summarized text

TABLE 6.2.3 ATTACHMENT PROCESSING TEST

Test Case ID	Test Case Name	Purpose	Input	Output
1	Attachment Reading	To check whether the system extracts text from attachments	PDF/DOC attachment	Extracted text from document

TABLE 6.2.4 USER APPROVAL TEST

Test Case ID	Test Case Name	Purpose	Input	Output
1	Approval Check	To verify if the system waits for user approval before sending reply	User selects "Yes"	System proceeds to send email

TABLE 6.2.5 EMAIL RESPONSE TEST

Test Case ID	Test Case Name	Purpose	Input	Output
1	SMTP Email Sending	To check whether the system sends reply email successfully	Generated reply message	Email successfully sent

7. CONCLUSION & FUTURE ASPECTS

7.CONCLUSION & FUTURE ASPECTS

The project successfully demonstrates the development of an intelligent **Email Automation System** using **UiPath Robotic Process Automation (RPA)** integrated with **Generative Artificial Intelligence (AI)**. The system automates the process of retrieving emails, extracting content, summarizing information, and generating automated responses. Through effective workflow design and integration of AI-based summarization, the system significantly reduces manual effort required for managing large volumes of emails.

The implementation of user approval through **WhatsApp Web automation** ensures that automated responses remain accurate and reliable while still maintaining human control. The integration of scheduling options using switch case logic allows responses to be sent immediately or after a specified delay, improving flexibility in email communication management.

Overall, the proposed system improves efficiency, reduces response time, and enhances productivity by automating repetitive email-handling tasks. The project demonstrates how the combination of automation technologies and artificial intelligence can provide practical solutions for modern communication management.

7.1 PROJECT CONCLUSION

This project presents an automated solution for managing and responding to emails using **UiPath RPA and Generative AI technologies**. The system retrieves emails from the mail server using the **IMAP protocol**, processes the email content and attachments, and generates summarized information using AI-based techniques. The summarized content is sent to the user through **WhatsApp Web automation** for review and approval before sending the final response.

The use of automation significantly reduces the time required for reading and responding to emails while improving overall efficiency. By integrating AI-based summarization with RPA-based automation workflows, the system provides an intelligent approach for handling email communication in organizations and businesses.

The successful implementation of email retrieval, summarization, user approval, and automated reply generation proves the effectiveness of the proposed system. This solution can be applied in various domains such as customer support, corporate communication, and administrative email management.

7.2 FUTURE ASPECTS

Although the proposed system performs effectively in automating email management, there are several opportunities for future improvements and enhancements.

Some potential future developments include:

•**Advanced AI Integration:** Incorporating more advanced AI models to improve email understanding and response generation.

•**Voice-Based Interaction:** Allowing users to approve or respond to emails using voice commands.

•**Priority-Based Email Classification:** Automatically categorizing emails based on urgency and importance.

•**Integration with Enterprise Systems:** Connecting the system with CRM and business management platforms for improved workflow automation.

•**Mobile Application Integration:** Developing a mobile interface for easier monitoring and approval of automated responses.

•**Enhanced Security Mechanisms:** Implementing advanced security measures to ensure safe handling of sensitive email information.

These future enhancements will further improve the system’s efficiency, scalability, and usability, making it a more powerful solution for automated communication management.

8. BIBLIOGRAPHY

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[2] Karpagavinayagam S., Sowndharya M., “AI-Powered Email Automation for Customer Query Handling in UiPath,” International Journal of Innovative Research in Electrical, Electronics, Instrumentation and Control Engineering (IJIREEICE), 2025.

[3] Kumar, N., Reddy, S., & Joshi, V.,“AI-Powered Email Automation System Using UiPath,”Proceedings of the International Conference on Artificial Intelligence and Automation.

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[5] Singh, A., Verma, K., & Patel, M.,“RPA-Driven Email Automation for Spam Detection and Classification,”International Journal of Advanced Research in Computer Science.

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8.2 GITHUB LINK

https://github.com/Ganesh-8639/Email-Automation-UiPath/blob/main/Main.xaml